

followed. Click 'Continue' button (refer Fig. 2).

3. Select a chapter of your choice using a pull-down menu (refer Fig. 3).

4. Start attempting the questions (refer Fig. 4). The questions here are generated randomly from the database stored in the server machine. You can select the respective radio button for your choice of answer.

5. As soon as you select an option, click 'submit' button to submit the answer and move to the next question.

Note that chapters stored in the database can store any number of questions but from those questions, only 15 are displayed on the client side for each test.

At the end of 15 questions, a test report is generated (refer Fig. 5), where you can view your performance.

To view the correct answers after the test, just click 'Get Answers' button on the test report screen (refer Fig. 6).

Testing steps to run the program on a PC

Using a single PC, server and client programs can be run at the same time. Here, the server program is under the control of the administrator and the client program is accessed by the student. To achieve this, first the server program is made to run, followed by the client program as explained below:

1. Download 'Java SE Development Kit (JDK) 6' from 'http://java.sun.com/javase/downloads/index.jsp' website.

2. Install JDK in your machine. By default it is installed under 'c:\program files\java\jdk1.6.0\bin' directory.

3. Install MS Access application program and configure the data source for accessing MS Access database from Java as follows:

- Open 'Administrative Tools' from 'Control Panel.'
- Now open 'Data Sources (ODBC)' and click 'Add' button.
- Locate 'Microsoft Access Driver (*.mdb)' and click 'Finish.'

Number of Question(s)	15
Number of Question(s) attempted	15
Number of Question(s) not attempted	0
Total Marks allocated	15
Total Marks obtained	6
Total Percentage	40.0%

Get Answers

Fig. 5: Test report

(d) Type 'Data Source Name' as 'OnlineExams' while observing case sensitivity.

(e) Assuming that all the relevant programs of this project are stored under 'D:\Rahul' folder, locate the database stored in 'D:\Rahul\' by using 'Select' button of the same window. When the database is selected, you will see the path of your database under 'Database' of the same window.

(f) Click 'Ok' to complete the configurations.

4. Run the server program. All the questions are stored in 'Questions And Answers' file of 'Rahul' folder. Open the command prompt from 'Start\Run...' option and type 'cmd' from the keyboard. Open 'Rahul' folder from 'D' drive. Type the appropriate path against the prompt as given below:

```
*Path = c:\program files\java\
jdk1.6.0\bin"
```

This is the path where the JDK files like javac, java and appletviewer are located.

5. Now type command 'javac ServerOnline.java' to compile the ServerOnline.java program. If you have set the path correctly, you will find class files under 'D:\Rahul.' Else, you will find error. Ignore warnings, if any.

6. Next type 'Java ServerOnline' on first command prompt to execute the software. You will see "Server Started !!!" in the next line.

7. Minimise the window and then

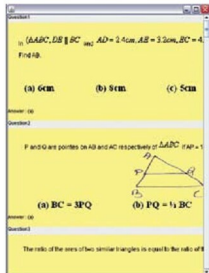


Fig. 6: Solutions to the questions

go to the next step to run the client program.

The steps for compiling and running the client program are the same as mentioned above. That is, click 'Start/Run' option followed by 'cmd.' Open 'D:\Rahul' folder and give the path of the JDK files.

8. Now instead of 'ServerOnline.java,' type 'ClientOnline.java' against the prompt.

If you run the ClientOnline program successfully, you will see the screen as shown in Fig. 1. A PC can handle multiple clients, so you can minimise the current client operation and open another client program on the same machine.

Testing steps to run on two PCs

1. Make sure that the PCs used for the project are connected on a local area network (LAN).

2. Place 'Rahul' folder in server and client machines. This is mandatory. Next, follow Steps 1 through 3 under sub-heading "Testing steps to run the program on a single PC."

3. Provide an appropriate path for JDK as mentioned in Step 4 of the above sub-heading. Compile the ServerOnline.java program and run it on the server machine.

4. On the client PC, the Connect-